

4 Correlation and regression

What students need to learn:

Scatter diagrams. Linear regression.

Calculation of the equation of a linear regression line using the method of least squares. Students may be required to draw this regression line on a scatter diagram.

Explanatory (independent) and response (dependent) variables. Applications and interpretations.

Use to make predictions within the range of values of the explanatory variable and the dangers of extrapolation. Derivations will not be required. Variables other than x and y may be used. Linear change of variable may be required.

The product moment correlation coefficient, its use, interpretation and limitations.

Derivations and tests of significance will not be required.
